

Reconfigurable LFSR Self-Testing for Three-Dimensional Stacked Integrated Circuit

Tian Chen, Ziyuan Fan, Beibei Luo, Jun Liu, Xiaohui Yuan, Huaguo Liang, Yingchun Lu

Abstract—Existing IC tests face challenges of decreased testability and increased costs of Three-Dimensional Stacked Integrated Circuit (3D-SIC) and changes in the test flow. This paper proposes a low-cost built-in self-test structure based on reconfigurable Three-Dimensional Linear Feedback Shift Registers (LFSR). In pre-bond tests, LFSRs on each layer independently generate test patterns. In mid-bond and post-bond tests, the LFSRs of each layer, before bonding, are reused to connect the LFSRs of multiple layers and form a larger LFSR, thereby generating test patterns. Our method applies to all pre-bond, mid-bond, and post-bond tests, reducing the total area overhead. The structure complies with test standards and supports serial-parallel hybrid tests. Additionally, we present a test data compression method that combines threshold-compatibility compression with graded LFSR encoding compression. The compatibility threshold is set in the test pattern compatibility process to control the maximum number of deterministic bits of the test patterns after compatibility within the threshold range. The test patterns after compatibility are divided into multiple classes according to the number of deterministic bits. Different-sized LFSRs are constructed for each class to encode the test patterns into seeds of different lengths. The proposed test data compression method effectively reduces the amount of test data storage and test time. Our experimental results on the ISCAS'89 circuit indicate that, compared to non-reconfigurable BIST methods, the proposed method reduces test area overhead by 44.71% and reduces test data storage and test time by 98.31% and 82.35%, respectively.

Index Terms—reconfigurable Three-Dimensional LFSR; test data compression; 1838; IEEE standard; 3D-SIC test; BIST.

I. INTRODUCTION

The size of interconnect wires of integrated circuits, especially global wires, has been increasing with the size of semiconductor devices. The integration of hundreds of millions of transistors on a single two-dimensional integrated circuit is approaching its physical limits [1]. Three-Dimensional Stacked Integrated Circuit (3D-SIC) technology is considered an effective way to overcome interconnect bottlenecks [2]. Recent research on monolithic three-dimensional integrated

circuits (M3D ICs) further indicates that three-dimensional connections should be explicitly incorporated into the physical design flow [3], which highlights the importance of vertical interconnect design in 3D/M3D ICs. While 3D structures significantly improve die performance, they bring many new challenges, such as heat, yield, and test issues during fabrication. Among them, 3D-SIC testing is an essential part of 3D dies, from theoretical research to product implementation [4]–[6]. The key factors of the test cost are the amount of test data storage, test time, and area overhead [7], which is the main factor restricting the overall profitability of the die. Therefore, minimizing the test cost is a vital issue of the semiconductor industry [8].

In two-dimensional integrated circuits, the large amount of test data generated by traditional external test methods must be loaded into the circuit under test by automated test equipment (ATE). The test cost is very high [9]. A more critical problem is that the ATE test machines require contact between the probes on the probe card and the solder pads or bumps on the die to establish a connection and achieve the test objectives. However, in the stacked structure of 3D-SIC, as the size of the bumps decreases, the number of bumps increases, and the spacing between the stacked layers is small. As a result, traditional ATE requires certain specifications for probe diameter, length, and other parameters, necessitating more miniature probes. Additionally, in 3D stacking manufacturing, each layer of dies undergoes thinning processes, so the pressure applied during probe tests may cause irreversible damage to the dies. Built-In Self-Test (BIST) is used to reduce the dependence of the die test process on ATE.

BIST is a design for testability (DFT) technology that is applicable to almost all types of integrated circuits. BIST embeds the test pattern generator, test controller, and response analyzer in the die, which can minimize the hardware and lower the cost of the self-test circuit [10]. Linear Feedback Shift Register (LFSR), as the most widely used BIST test pattern generator [11], is used in pseudo-random and deterministic tests. The 2-D LFSRs generate both precomputed test patterns (for detecting random-pattern-resistant faults) and random patterns (for detecting random-pattern-detectable faults) and have the advantages of high fault coverage and at-speed testing [12]. The method of generating precomputed deterministic test patterns using an LFSR is also referred to as LFSR reseeding [13]. Its idea is to encode the precomputed deterministic test vectors into smaller test seeds and store them. During the chip testing application process, the LFSR is used to expand the seeds into the required sequence of deterministic test patterns. Since the seed length is much shorter than the deterministic test

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patterns, the test data storage volume is significantly reduced. Hellebrand [14] proposed a multiple polynomial LFSR (MP-LFSR) reseeding method, where the additional gate arrays of AND make the structure of the LFSR controllable. With 16 polynomials, a shorter seed can achieve the same encoding effect [11]. Variable-length seed LFSR [15] reduces the coding length and lowers the area overhead. Sun [16] presented a method that combines dictionary coding and partial LFSR reseeding to improve the compression efficiency of test data compression. However, since it requires decompression of the dictionary-encoded portion, it increases hardware overhead. Alamgir [17] proposed a parallel LFSR reseeding scheme (C-LFSR), which uses one N_1 -stage LFSR with N_1 and two N_2 -stage LFSRs to generate test patterns jointly. The test patterns generated by the N_1 -stage LFSR are used to control the test patterns generated by the N_2 -stage LFSR. These three LFSRs are smaller and their sums of stages are less than the original LFSR's number of stages, which reduces the number of storage elements in the LFSR and thus reduces the hardware overhead. Pomeranz [18] filled the seed differently to obtain a new seed with a longer bit length. This method needs to store the original seed and the additional filling bits to detect more target faults with less test data.

Existing studies have proposed BIST test architectures for 3D-SICs to reduce hardware overhead and improve test efficiency. Alamgir [19] designed a distributed and reusable BIST controller architecture, in which the BIST controller used in the pre-bond test stage can also be reused in subsequent test stages. However, this method does not consider the reuse of the test pattern generator. Chen [20] proposed a low-power LFSR structure for testing three-layer stacked chips. In the mid-bond or post-bond test stage, part of the LFSR structure used in the pre-bond test is employed as the main LFSR, while the remaining part is used as the Mask-LFSR. A dual-LFSR structure is then used to generate low-power test patterns. This structure is suitable for three-layer stacked chips, but it is not applicable to stacked structures with more layers.

In addition to the above 3D-SIC BIST methods, recent studies on M3D ICs have also focused on BIST solutions for high-density and realistic ILV layouts. Chaudhuri et al. [21] proposed a BIST framework for high-density and realistic ILV layouts in M3D ICs, in which irregular ILV distributions and practical layout constraints are considered for ILV fault detection and localization. Jahanirad et al. [22] proposed a cluster-based BIST architecture for M3D ICs, where ILVs are divided into different clusters according to fault occurrence characteristics, and a walking-pattern-based test strategy is adopted to improve fault coverage and approximate localization capability. These methods can effectively address dedicated ILV fault detection and localization in M3D ICs. Chen et al. [23] proposed a scan-chain-based ILV BIST architecture, which can detect open faults, stuck-at faults, and all possible shorts under irregular ILV layouts. This method also uses a k-means clustering algorithm to group ILVs for testing, significantly reducing test hardware overhead while achieving accurate fault localization.

Despite these advances, it should be noted that ILV-based monolithic 3D integration and TSV-based 3D stacking are

fundamentally different in terms of fabrication process and physical design constraints. Specifically, ILVs are formed during sequential integration, whereas TSVs are fabricated before bonding. Therefore, optimization methods developed for M3D architectures are not directly applicable to TSV-based 3D stacked integrated circuits. In addition, TSV-based 3D stacking remains a mainstream technology in current industrial practice due to its mature fabrication process, chip-level test flexibility, and heterogeneous integration capability. Therefore, our work continues to focus on TSV-based 3D stacked integrated circuits, and addresses specific test challenges in multi-layer stacking, including the reuse of test hardware across bonding stages and efficient test data compression for large-scale stacked circuits.

In LFSR reseeding methods, the critical issue for achieving complete fault coverage lies in whether all precomputed deterministic test vectors can be successfully encoded into seeds. If a test vector cannot be encoded into an LFSR seed by an LFSR with a specific primitive polynomial, it may still be encoded into a seed by another LFSR with a primitive polynomial of the same or a higher number of stages. The size of the LFSR is related to the maximum number of deterministic bits in the test vectors. The LFSR-based BIST needs to minimize the number of stages in the LFSR. Consequently, it is often difficult to achieve complete fault coverage during testing.

3D-SIC features a large circuit scale, substantial test data volume, and a complex test process. Directly applying LFSR-based BIST methods to 3D-SIC testing leads to significant cost issues. Given an n -layer 3D-SIC, the testing needs to be performed separately for each layer. A pre-bonding test is conducted for each layer during fabrication, adding up to n tests. During the chip stacking process, mid-bond testing is required to ensure that each component involved in the bonding is a known good die (KGD), with a total of $n - 1$ tests needed. Lastly, a post-bond test is implemented before the completion of stacking and packaging. Therefore, an n -layer 3D-SIC needs to be tested $2n$ tests. In contrast, testing n non-stacked 2D dies requires only n tests since no bonding process is involved. That is, a 3D-SIC requires twice as many tests as conventional 2D dies containing the same number of dies. The 3D stacking process must be performed between KGDs, which necessitates that each die and partially stacked die should be designed to be testable. Due to the different under-test objects in each testing round, the required Design-for-Testability (DFT) circuit structures may also vary. These DFT circuits cannot be added temporarily during the stacking process; instead, they are embedded in each layer before bonding. As the number of layers in 3D SICs increases, if DFT circuits cannot be effectively reused across multiple stacking stages, the total overhead of DFT circuits required for 3D SICs will increase greatly.

Since the control signals and test data for the upper-layer die in a stacked configuration can come from the lower-layer die through Through-Silicon Vias (TSVs), how to manage, coordinate, and schedule the test patterns on different dies and cores in a multi-layer 3D-SIC is a significant issue to maximize the reduction of test time. The IEEE Standards Association published the 3D-SIC DFT standard, IEEE Standard 1838-

2019 [24] that defines the test flow and modes for each component (stacked dies, signal pins, signal channels, etc.) in a 3D-SIC and tests the connectivity and reliability between components in a 3D-SIC. It also defines a mandatory serial access mechanism and an optional flexible parallel access mechanism, which can well support the scheduling of inter-layer serial or parallel tests. The scheduling is achieved through predefined test modes to achieve comprehensive and efficient test coverage. Some 1838-compliant 3D-SIC design architectures have been developed [25]–[27]. At the same time, the interconnect test access mechanism between cores within a layer needs to be emphasized. Test access architecture standards for 3D-stacked ICs are conceptually compatible with previously developed 2D-IC DFT standards, in particular IEEE Std 1149.1 [28], IEEE Std 1500 [29], and IEEE Std 1687 [30]. The first two standards specify the test access architecture for on-board ICs and IP cores embedded within ICs, and IEEE Std 1687 describes the access architecture for embedded instrumentation. Combining these four standards enables the interconnection to the 3D-SIC interlayer and intralayer cores.

This paper proposes a reconfigurable R-BIST architecture that applies to 3D-SIC structures with an arbitrary number of layers. In the pre-bond test, the LFSRs of each layer in this architecture are combined in the mid-bond and post-bond tests to reconstruct a larger LFSR. When all LFSRs of all layers are connected simultaneously, the LFSR has the most significant scale and the strongest coding capability. Furthermore, this structure supports the reuse of controllers in different test phases.

In the R-BIST architecture, LFSR seeds are stored in on-chip ROM or off-chip. The storage capacity of seed data is one of the key factors influencing test cost. To mitigate the substantial growth of test data storage, we propose a data compression method, which applies threshold-compatibility compression to test patterns, reducing the number of test patterns without increasing the maximum number of deterministic bits. The compressed test patterns are divided into multiple classes based on the number of included deterministic bits, and the test patterns in each class are encoded using different sizes of LFSR. The R-BIST structure is responsible for implementing the LFSR structure at different scales. As a result, the original test set is compressed into seeds of varying lengths, effectively reducing the amount of test data storage. The main difference between this method and traditional test data compression methods is that this method is the first test data compression method designed according to the characteristics of 3D-SIC test structures.

The proposed R-BIST architecture is compatible not only with IEEE Std 1838 but also with 1149.1, 1500, and 1687 standards. Leveraging multiple test standards enables the implementation of serial-parallel mixed tests among different layers of 3D-SIC and cores within any layer. Test patterns are compactly delivered into the circuits under test, which reduces the test interval and, thus, effectively reduces the test time.

The main contributions of this paper are as follows:

- 1) This paper proposes an R-BIST architecture based on a reconfigurable three-dimensional LFSR. The proposed architecture can reuse LFSR resources across different

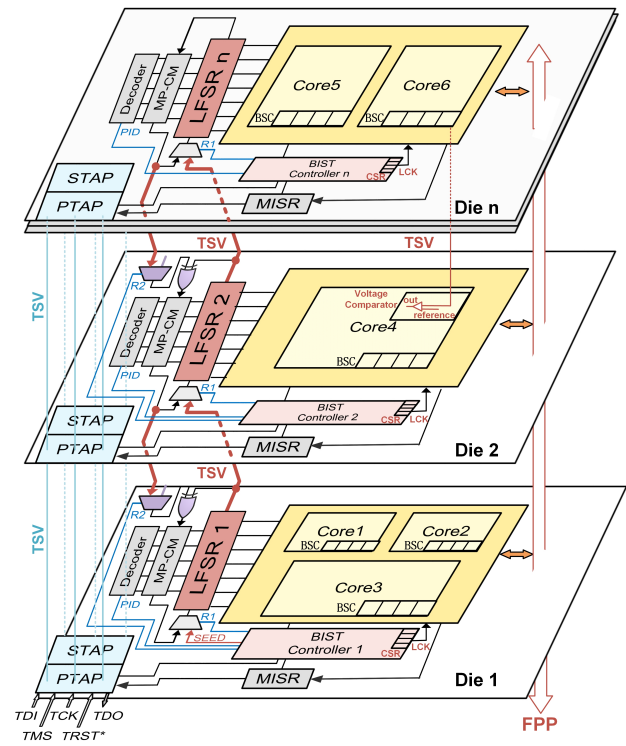


Fig. 1. Overview of the R-BIST structure.

test stages. Specifically, the LFSRs used in the pre-bond test stage can be reused in the mid-bond and post-bond test stages, thereby reducing additional test hardware overhead. Meanwhile, the proposed R-BIST architecture is compatible with the 3D-SIC test standard IEEE Std 1838, as well as the 2D IC test standards IEEE 1149.1, IEEE 1500, and IEEE 1687, thus supporting both inter-layer and intra-layer test access mechanisms. In addition, the architecture supports a serial-parallel hybrid test mode, which enables flexible scheduling of the test process for different layers and cores according to test requirements, thereby shortening the overall test time.

- 2) A threshold-compatibility compression with a graded LFSR encoding compression method for 3D-SIC test patterns is proposed, and reduces test data storage.

The rest of this paper is organized as follows: Section II presents the details of the reconfigurable R-BIST framework, Section III describes the new test data compression method, IV discusses the experimental results. Section V concludes this paper with a summary.

II. RECONFIGURABLE BUILT-IN-SELF-TEST

The proposed Reconfigurable Built-In-Self-Test (R-BIST) architecture is decentralized and flexible for testing 3D-SIC. The proposed n -layer die test architecture is illustrated in Fig. 1. The architecture consists of Linear Feedback Shift Register (LFSR), Multiple-Polynomial Control Model (MP-CM), Decoder, BIST Controller, Multiple-Input Signature Register (MISR), Primary Test Access Port (PTAP), Secondary Test

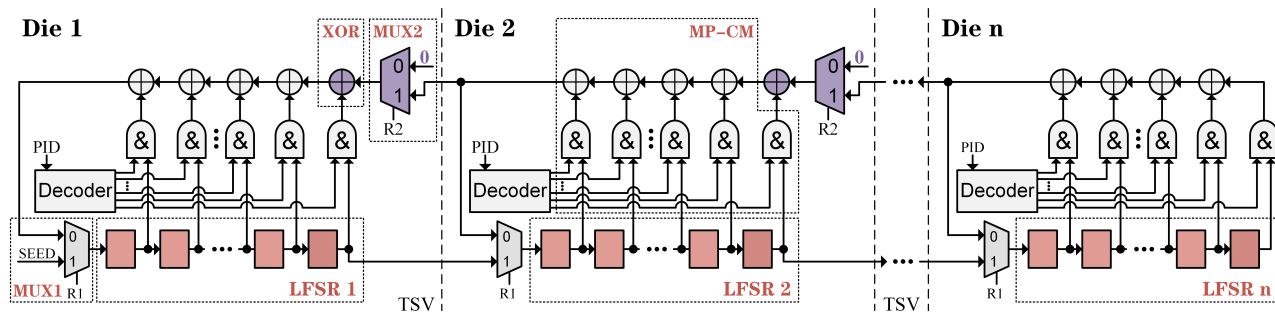


Fig. 2. The structure of an n -layer reconfigurable LFSR.

Access Port (STAP), Inter-Layer Flexible Parallel Port (FPP), and logic gates, e.g., XOR and AND, and two-way selectors.

Test control signals are loaded through the PTAP on the bottom-layer chip, and the bottom-layer BIST controller 1 transfers the control signals and seed data to the appropriate layers and components. The Decoder controls the MP-CM to adjust the XOR feedback network of the LFSR, enabling the reconfiguration of LFSR structures with different polynomials, and the LFSR loads test seeds. The lower-layer BIST controller transmits control signals and data to the upper-layer PTAP via STAP and TSV, while the upper-layer BIST controller parses the control signals and data obtained from the local-layer PTAP. Through the BIST controller and Test Access Port, test scheduling for each layer can be achieved. The reconfigurable LFSR structure enables flexible configuration of LFSRs in each layer according to test requirements, satisfying the generation conditions of different test vectors. Each layer's BIST controller is configured with a Circular Shift Register (CSR). Test vectors are scheduled among the layers based on the clock cycles allocated by the circular shift register group, thereby reaching the designated cores. Finally, the test responses are compressed by MISR into test features, which are compared with the characteristics of known fault-free circuits to ensure the circuit's correctness.

A. Reconfigurable LFSR

The R-BIST architecture has an R3D-LFSR at its core, which adjusts the test pattern generator architecture based on the requirements for different stages of 3D ICs. A single-stage LFSR is used as the test pattern generator when the test patterns have few deterministic bits. When the test patterns have more deterministic bits, a multi-layer LFSR structure is used to cascade and construct an LFSR with many layers. This helps to improve the ability to generate test patterns. Fig. 2 displays the connection structure of the n -layer reconfigurable LFSR. The structure consists of Die 1, Die 2, and Die n . Each layer of Die communicates with others utilizing TSVs.

In each layer's LFSR, a two-way selector MUX1 is configured before the first D flip-flop, and its control signal is R1. When the control signal R1 is set to 1, the seed is shifted in the LFSR; when the control signal R1 is set to 0, the LFSR connects to the feedback network to generate the test pattern. A two-way selector MUX2 and a two-input XOR gate are attached to the end of the LFSR for each layer except layer

n . MUX2's control signal is R2, and one of its inputs is fixed to 0. When R2 is set to 0, MUX2 outputs 0; when R2 is set to 1, MUX2 outputs the value of the feedback data from the previous layer of LFSRs. XOR is directly connected to the output of MUX2. The other input value of XOR comes from the last flip-flop of the current layer of LFSRs. MUX2 and XOR perform selective multiplexing of LFSRs between different layers. When R2 is 0, the output of MUX2 is 0, and XORing any value with 0 does not change that value. Thus, the output value of XOR equals the output value of the final flip-flop of this layer. The LFSR of this layer independently generates test patterns unrelated to the upper layer's LFSR. When R2 is 1, the last flip-flop of the current layer is combined with the feedback data network from the upper layer to form a larger XOR feedback network. The upper and lower layers of the LFSRs are connected, constituting an LFSR with a larger scale that jointly participates in generating test patterns. The decoder is controlled by the signal PID. Each AND gate in the Multiple-Polynomial Control Model MP-CM can be turned on or off by the decoder to change the polynomials of the LFSRs. Overall, the control signals such as R1, R2, and PID are appropriately set according to the test requirements, and the LFSRs of each layer are configured rationally. It is possible to use only a single-layer LFSR for testing or co-configure two-, three-, or n -layer LFSRs to form a larger LFSR with superior ability to generate test patterns. This addresses the issues of large test data volume, numerous deterministic bits in test vectors, and difficulty in seed generation for 3D integrated circuits. Without additional LFSR flip-flop structures, the R3D-LFSR can be used to generate more test data for stacked 3D integrated circuits, which is applicable not only to the testing of circuit nodes but also to the testing and localization of interconnections such as TSVs.

Fig. 3 gives a simplified diagram of the reconfigurable LFSR and two test pattern generation paths. In the first path, labeled as ①, $LFSR1$ is used to generate test patterns. Both R1 and R2 in Die 1 are set to 0. This path suits pre-bond tests or tests with fewer test pattern determinants during/after bonding. Path ② spans two layers in this configuration, with R1 set to 0 in Die 1 and Die 2, R2 set to 1 in Die 1, and R2 set to 0 in Die 2. At this stage, the interlayer TSVs connect $LFSR1$ on Die 1 and $LFSR2$ on Die 2, and the sum of the stages of both LFSRs becomes the resulting LFSR stages. A longer LFSR can be utilized to generate test patterns with more

is

$$P(X_{t+1} = d | X_t = d) = \begin{cases} \frac{1}{2} \cdot \frac{2^d - t - 1}{2^k - t - 1}, & (d > 0, t + 1 \leq 2^d - 1) \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

$$P(X_{t+1} = d + 1 | X_t = d) = \begin{cases} \frac{2^k - 2^d}{2^k - t - 1}, & (0 < d \leq k - 1) \\ 0, & \text{otherwise.} \end{cases}$$

The coding success rate is

$$P_{seed}(k, 0, s) = \sum_{d=1}^s P(X_s = d) = \sum_{d=\lceil \log_2(s+1) \rceil}^{\min(k,s)} P(X_s = d). \quad (2)$$

The probabilities $P(X_s = d)$ can be determined recursively using $P(X_t = d) = P(X_{t-1} = d) \cdot P(X_t = d | X_{t-1} = d) + P(X_{t-1} = d - 1) \cdot P(X_t = d | X_{t-1} = d - 1)$.

Corollary. Let $C \in \{0, 1, x\}^m$ be a test pattern of length m , and suppose that C contains s deterministic bits. For a k -stage LFSR with 2^q candidate characteristic polynomials, if the probability that C can be successfully encoded into an LFSR seed using a single characteristic polynomial is $P_{seed}(k, 0, s)$, then the probability that C cannot be successfully encoded into a seed under all 2^q characteristic polynomials is

$$P_{noseed}(k, q, s) = [1 - P_{seed}(k, 0, s)]^{2^q}. \quad (3)$$

Proof. Let A_i denote the event that the test pattern C can be successfully encoded into an LFSR seed using the i -th characteristic polynomial, where

$$i = 1, 2, \dots, 2^q. \quad (4)$$

According to the LFSR coding theory under a single characteristic polynomial, we have

$$P(A_i) = P_{seed}(k, 0, s). \quad (5)$$

Therefore, the event that C cannot be successfully encoded into a seed using the i -th characteristic polynomial is A_i^c , and its probability is

$$P(A_i^c) = 1 - P(A_i) = 1 - P_{seed}(k, 0, s). \quad (6)$$

Since the encoding processes under different characteristic polynomials can be regarded as independent and identically distributed events, $A_1, A_2, \dots, A_{2^q}$ are independent and identically distributed. The failure of encoding C using a k -stage LFSR with 2^q characteristic polynomials is equivalent to the event that C fails to be encoded under all 2^q characteristic polynomials, namely,

$$A_1^c \cap A_2^c \cap \dots \cap A_{2^q}^c. \quad (7)$$

According to the multiplication rule of independent events, we obtain

$$P_{noseed}(k, q, s) = P(A_1^c \cap A_2^c \cap \dots \cap A_{2^q}^c) = \prod_{i=1}^{2^q} P(A_i^c) = \prod_{i=1}^{2^q} [1 - P_{seed}(k, 0, s)] = [1 - P_{seed}(k, 0, s)]^{2^q}. \quad (8)$$

Therefore, when the number of candidate characteristic polynomials is 2^q , the probability that the test pattern C cannot be successfully encoded into an LFSR seed is given by Eq.(3).

Table I gives the coding failure rate $P_{noseed}(k, q, s)$ for each representative value of k, q and s . The solid and dashed lines in Fig. 5 represent the coding failure rates at $k=20$ and $k=40$, respectively. At point **A** there is $P_{noseed}(20, 0, 16) \approx 0.031$ and at point **B** there is $P_{noseed}(20, 1, 16) \approx 0.001$, i.e., the coding failure rate decreases by a factor of about 30 when $k-s=4$ and q increases by 1. When $k-s=10$ and q increases by 1, **A'** and **B'** differ by a factor of about 2,000. That is, the larger the difference in $k-s$, the more significant the increase in the number of polynomials on the coding success rate. At point **C**, there is $P_{noseed}(40, 0, 16) \approx 2.97 \times 10^{-8}$, which increases the coding success rate by about 10^6 times compared to point **A** with q and s unchanged and k increased to 40. Increasing q under the conditions of point **C**, we have $P_{noseed}(40, 1, 16) \approx 8.84 \times 10^{-16}$ at point **D**. Compared to point **A**, s is unchanged, both k and q are increased, and the encoding success rate is improved by a factor of about 10^{12} ($> 3 \times 10^7$). Simultaneously increasing the number of polynomials and the number of LFSR 'stages can bring about an increase in the coding success rate and bring about a gain that is superior to the superposition of the two implemented separately.

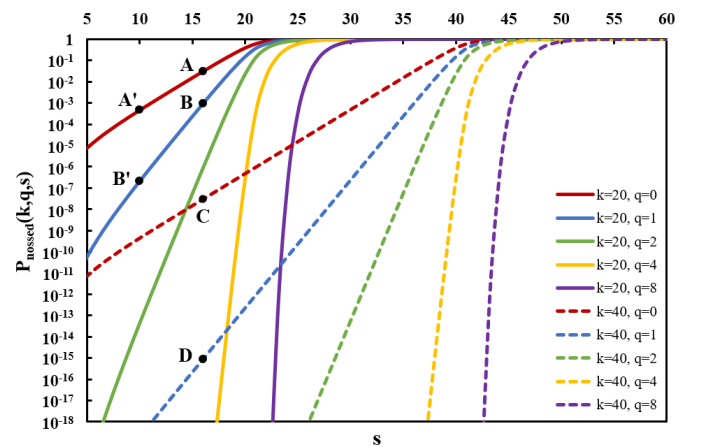


Fig. 5. Theoretical values for $P_{noseed}(k, q, s)$ when $k=20$ and $k=40$.

Suppose an n -layer 3D-SIC has a k_i -stage LFSR containing 2^{q_i} polynomials on its i -th layer ($1 \leq i \leq n$). The proposed reconfigurable LFSR structure multiplexes multiple layers of

LFSRs, and the lowest coding failure rate achievable for n -layer 3D-SIC is $P_{noseed}(k_1 + k_2 + \dots + k_n, q_1 + q_2 + \dots + q_n, s)$. Since the number of polynomials available after the cascading of LFSRs is $2^{q_1 + q_2 + \dots + q_n}$, it is not just a mere add, but increases exponentially, and at the same time the number of LFSR cascades increases linearly, the proposed reconfigurable LFSR structure improves the coding capability significantly and facilitates the test pattern seed solution for large circuits.

C. Mixed Serial-parallel Test Module

The test patterns generated by the R3D-LFSR need to be sent to the scan chains to be tested. Multiple scan chains on different layers or cores may share the same test pattern. In serial test mode, the test pattern is sent to the first scan chain, then to the second scan chain after all the test patterns are sent, and so on, until all the scan chains sharing the test pattern are filled. Finally, the test is initiated, and the test response is captured.

To improve test efficiency and shorten test time, parallel test mode is supported, allowing one test pattern to be fed into multiple scan chains in different layers or cores simultaneously. The 3D-SIC DFT standard IEEE 1838, published in 2019, defines PTAP (Primary Test Access Port), STAP (Secondary Test Access Port), and FPP (Flexible Test Access Port). For implementing test scheduling, cyclic shift registers (CSR) are set up in each layer's Built-In Self-Test (BIST) controller to allocate different time slots for core testing within each layer.

To enhance test efficiency and achieve test scheduling, a cyclic shift register is configured in the BIST controller of each layer to allocate different clock cycles for the core testing of each layer. The core circuits under test in different layers will obtain the test vectors of the test circuits from the FPP channels within the clock cycles of each layer, and the test vectors will be stored in the scan chains of the core circuits under test. The R-BIST framework incorporates the IEEE 1838 structure and is proposed to enable efficient serial-parallel tests, thereby improving test efficiency.

Fig. 6 depicts the test architecture on the second and third layer of the die in detail, which contains the three cores to be tested. The dashed-line box represents a reconfigurable LFSR, which has been extensively described in the preceding sections. The focus shifts to other components used to implement serial or parallel tests. In the R-BIST architecture, an inter-layer cyclic shift register (CSR) with the same structure but different initial values is configured in each layer of the stacked BIST Controller. This register decomposes the global test clock GTC (Global Test Clock) signal into intra-layer test clocks LTC (Layer Test Clock). The decomposed GTC is reduced to a lower level, serving as the test clock for the cores. When a test vector needs to be simultaneously sent to multiple scan chains in different layers or different cores, the same clock cycle is allocated to the different layer cores requiring this test vector, enabling the scheduling of test cores in different layers to retrieve test vectors from the FPP. Cores receive data from the shared FPP channel based on the test clock, thereby ensuring that each core can perform time-division parallel testing at a lower frequency.

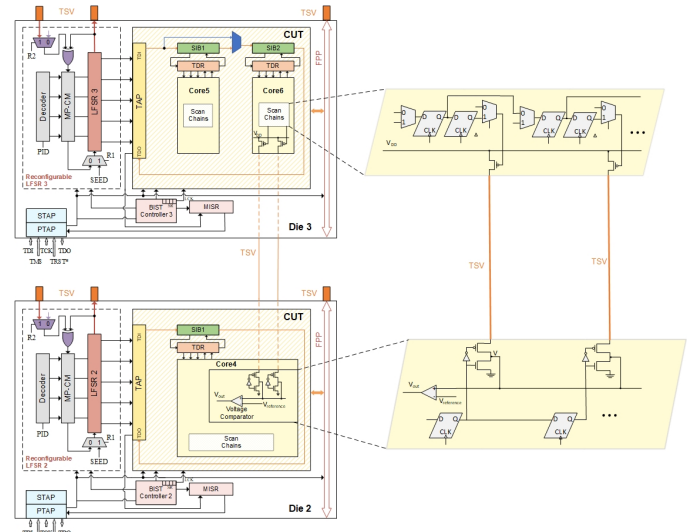


Fig. 6. The structure of layers two and three in detail.

Both the STAP and PTAP ports are defined by the 1838 standard. PTAP is a mandatory test port allowing serial access to all test and debug functions on the die. It is a TAP based on JTAG (IEEE 1149.1). The test interface includes test input TDI, clock TCK, test data output TDO, mode selection TMS, and an optional test reset TRST*. Control commands, such as initiate test, stop test, and reset, are sent to the BIST controller through the PTAP. The test port STAP is connected to PTAP in the upper layer to propagate signals related to the upper die test, serving the function of signal propagation. PTAP and STAP provide a mechanism for transmitting test data and test control signals into the inside of the die. They also enable the transmission of signals from one layer to the upward or downward layer, providing a mechanism for inter-layer serial access.

The Flexible Parallel Port (FPP), which meets the 1838 standard, establishes a connection between PTAP, STAP, and the core, providing a parallel test access interface. FPP consists of a set of lanes, each of which implements a 1-bit-wide data channel that can be unidirectional or bidirectional. The FPP transmits multiple test data bits at one time and is a high-speed data port that transmits test data in parallel between the upper and lower layers of the die. The cores of the circuit under test (CUT) in each layer will retrieve test data from the FPP within the allocated LTC (Layer Test Clock) cycles, and the test data will be stored in the scan chains of each core. Within the LTC cycles of each layer, the cores on the corresponding die can exclusively occupy the communication channel to transmit or receive data. During the testing process, inter-layer test scheduling is achieved by allocating LTCs for different layers: associated dies undergoing parallel testing will be assigned to the same LTC, while those undergoing serial testing will be allocated to different LTCs. The above components assist in accomplishing the interlayer serial-parallel tests of 3D-SIC. Other standard components are employed as a supplement for the inter-core serial-parallel tests of the Circuit Under Test containing multiple cores.

A TAP based on the IEEE 1500 standard is set up within the CUT. Test Data Registers (TDRs) and Segment Insertion Bits (SIBs) should be placed around the periphery of each core to enable testing. The TDR is based on the 1149.1 standard and is used to update and capture test data, while the SIB is based on the 1687 standard and is a unit that acts as a gate or switch to "segment" scan paths according to user requirements. When the switch (SIB bit) is set to OPEN, it permits adding scan paths to the JTAG network. When the switch is set to CLOSE, it excludes specific scan paths from the network, enabling bypass tests of the core and perfecting inter-core serial tests. In addition, by setting a multiplexer between two adjacent serial cores, the test data input from the TAP is broadcast to each core synchronously, realizing the inter-core parallel tests. The 3D-SIC test structure designed in this paper achieves efficient testing and fault detection by integrating components compliant with standards such as 1149.1, 1500, 1687, and 1838. It enables serial-parallel mixed test control for multiple cores located in different layers or within a single layer, which realizes efficient test and fault detection and saves test time.

D. Interlayer Interconnection TSV Testing

Our R3D-LFSR can be applied to core testing and to TSV (Through-Silicon Via) testing. This work adopts a voltage division testing method for TSV testing. Fig. 6 presents the TSV test structure. Testing a single TSV requires additional MOS transistors, D flip-flops, and a voltage comparator. The primary functions of the MOS transistors are to perform voltage division and control the switching of the TSV path under test. Taking the testing of a TSV between CORE6 and CORE4 as an example, one PMOS transistor is added for a TSV in Die3, while a combined MOS transistor voltage division structure composed of one PMOS, one NMOS, and an inverter is added in Die2. The D flip-flops used for TSV testing are connected to form a scan chain, which can store and shift TSV test patterns. These scan chains can reuse the boundary scan cells in core testing without requiring additional components. The voltage comparator is responsible for comparing the actual voltage value at the test point with the standard fault-free voltage value at the test point to determine whether a fault occurs in the TSV.

The testing principle is that test vectors required for TSV testing are generated by the LFSR (Linear Feedback Shift Register). These test vectors are shifted into the scan chain to control the conduction and cutoff of the MOS transistors. A voltage V_{DD} is applied to the TSV under test, and voltage division is performed using the resistances of the NMOS and PMOS transistors to obtain the output voltage of the TSV. The voltage value V at the midpoint between the NMOS and PMOS in Die2 is fed into one input terminal of the voltage comparator, while the other input terminal receives the voltage value $V_{reference}$ at this point under fault-free conditions for the TSV. The voltage comparator then compares the voltages at the two input terminals, outputs the test result V_{out} , and finally discriminates the faults in the TSV based on the value of V_{out} .

III. TEST DATA COMPRESSION

The test data volume for 3D-SIC pre-bonding, mid-bonding, and after-bonding tests is extremely large. To store and transmit this test data, a large amount of storage space and high transmission bandwidth are required, which increases the test cost. LFSR reseeding is an efficient test data compression method to tackle this issue. With this technique, only the seed of each test vector needs to be stored. It reduces the data storage and transmission time. Our testing method reduces the storage volume of seeds as shown in Fig. 7. First,

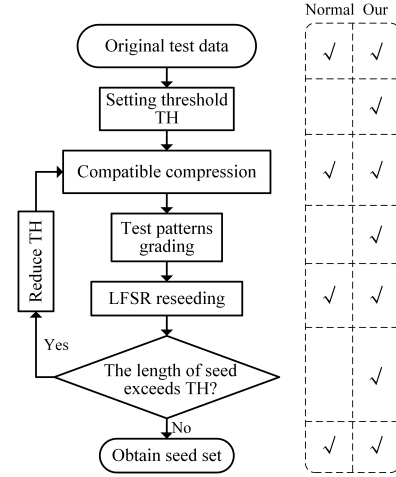


Fig. 7. Test data compression flowchart.

compatible compression preprocessing is performed on the original precomputed test vector set to reduce the number of test vectors, and then LFSR seeds are generated, which can decrease the number of seed vectors in the seed set. This process is illustrated in the "Normal" column of Fig. 7, representing the standard test data compression process.

The "Our" column in Fig. 7 denotes the steps (marked with check marks) of the data compression method. In our compression method, a threshold TH is added before the compatibility compression. After compatibility compression, there is a significant difference in the number of deterministic bits contained in each test vector, and the test vectors with a vast number of deterministic bits usually only account for a very small fraction of the total test vectors. The seed length of LFSR reseeding is positively correlated with a maximum number of deterministic bits (S_{max}), and compatibility compression may increase S_{max} in the test set. This makes it necessary to increase the number of stages of LFSR to ensure the probability of successful encoding, which leads to an increase in seed length. However, reseeding the entire test set with only these very few test vectors as standards, leading to a rise in seed length for test vectors containing a small number of deterministic bits, can result in a waste of storage space.

To address this issue, we set a threshold in the compatibility process for adjustment. If the S_{max} of a test vector after compatibility exceeds the pre-set threshold, the compatibility operation is rejected. In addition, a test vector grading operation is introduced after compatibility compression and before reseeding. Test vectors are divided into multiple classes

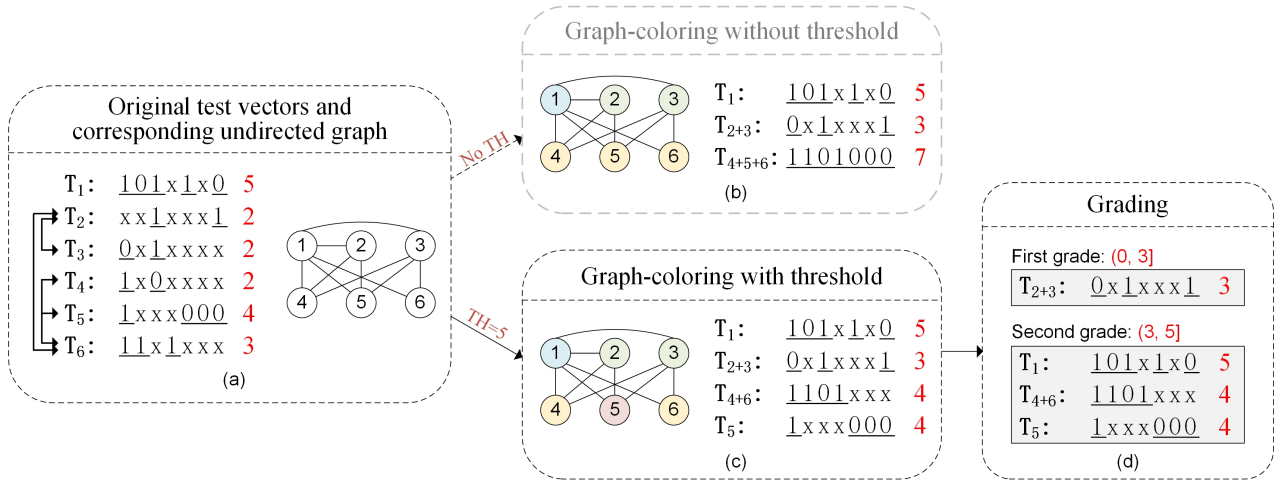


Fig. 8. Test data graph coloring compatibility, compression, and grading process. (a) Original test dataset. (b) Graph coloring compatibility compression results without threshold. (c) Graph coloring compatibility compression results with threshold (d) Grading.

based on the deterministic bits, with each class referred to as a grade. Test vectors with fewer deterministic bits are reseeded to generate short seeds using a short LFSR. In comparison, test vectors with more deterministic bits are reseeded to generate long seeds using a large LFSR that multiplexes multiple layers of LFSR. After reseeding, if the length of the generated seed exceeds the previously set threshold, the threshold is lowered to re-perform the compatibility compression. Examples of setting a threshold for compatibility compression and graded reseeding are illustrated below.

A. Threshold-compatibility compression

The test vector compatibility is that if for any specific position $a_i (1 \leq i < n)$ in the test vector $A = a_0a_1...a_n$, there is a corresponding position b_i in the test vector $B = b_0b_1...b_n$ such that a_i and b_i are the same, or one of them is a don't care bit x , then A and B are compatible, i.e., $A \equiv B$.

Before encoding into the LFSR seed set, compatible preprocessing needs to be performed on the precomputed original deterministic test set. This paper proposes an improved threshold-based graph-coloring compatible compression method. The graph-coloring compression method maps the test data and their relationships to an undirected graph structure. The undirected graph $G = (V, E)$ consists of vertices V and edges E . Each test vector is considered a vertex, and the compatibility relationship between test vectors represents an edge; an edge exists only when two test vectors are not incompatible. Fig. 8 (a) shows a set of mid-bond test vectors and their corresponding undirected graph. In $T_2 \equiv T_3$, $T_2 \equiv T_6$, $T_4 \equiv T_5 \equiv T_6$, each test vector is followed by the number of deterministic bits it contains, and S_{max} is 5.

The problem is formulated as coloring the vertices in a graph. The two vertices at the two ends of an edge cannot be of the same color, and the vertices of test vectors grouped in the same group are colored with the same color. The concept of the threshold-based heuristic graph-coloring algorithm is to select the next vertex to be colored based on the saturation degree and uncolored degree of the vertices. Saturation degree

refers to the number of colors used by the colored vertices adjacent to the current vertex. Initially, the saturation degree of all vertices is 0. A higher saturation degree indicates that more neighbors of the vertex have been colored, resulting in fewer available colors for the vertex. Therefore, prioritizing the selection of vertices with higher saturation degrees for coloring helps reduce subsequent coloring conflicts. Uncolored degree is defined as the number of uncolored vertices adjacent to a given vertex. If all uncolored vertices have the same saturation degree, vertices with higher uncolored degrees are selected, as these vertices have more uncolored neighbors and may require more colors. Processing them in advance helps reduce overall color usage. For each color, the number of vertices that have used it is tracked, and a color usage threshold TH is set. During the coloring operation, the number of times a color has been used is checked. If it exceeds the threshold TH , a new color will be used for coloring. In each iteration, the saturation degree and uncolored degree are updated based on the colored vertices. Finally, vertices colored with the same color correspond to test vectors grouped into the same set.

We obtain the graph-coloring results in Fig. 8 (b) based on the heuristic graph-coloring algorithm without setting the threshold. T_2 and T_3 are merged into the test vector T_{2+3} , and T_4 , T_5 , and T_6 are merged into T_{4+5+6} . After grouping and merging, S_{max} increases to 7. In practical situations, don't-care bits occupy the majority of the test vector bits. Patterns with a surge in the number of deterministic bits after compatibility exist, like from 17 to 30, but account for a tiny percentage of the patterns. Most of the test vectors after compatibility have an increase in the number of deterministic bits, yet the increment is slight, such as from 17 to 22.

The seed length of LFSR reseeding positively correlates with S_{max} , and the larger the S_{max} , the longer the seed is. It is not cost-effective to increase the seed length of other test vectors containing a small number of deterministic bits to satisfy the encoding requests of a very small number of patterns with a significant number of deterministic bits. Therefore, we set a threshold TH in the graph-coloring process, which constrains

the maximum number of deterministic bits of the test vectors after compatibility.

The graph-coloring result when TH is set to 5 is shown in Fig. 8 (c). Because the number of deterministic bits of T_{4+5+6} is 7, exceeding TH , they cannot be classified into the same group. The final graph-coloring result is that T_2 and T_3 are in the same group, T_4 and T_6 are in the same group, and T_1 and T_5 are grouped individually. In this example, S_{max} is 5, which is within the threshold. The LFSR stage without setting the threshold is 7, and there are 3 test vectors; the seed storage is $7 \times 3 = 21$ bits. After setting the threshold, the LFSR stage is 5; the number of test vectors is 4, and the seed storage is $5 \times 4 = 20$ bits. The graph-coloring strategy of setting the threshold increases the number of test vectors after compression, but reduces the data storage because it reduces the seed length. In the pre-bond test phase, the TH_i set for the die at layer i cannot be lower than the $S_{max,i}(TH_i \geq S_{max,i})$ of the original test cube. During the mid-bond and post-bond stages, as the number of LFSR stages for each layer is already determined, TH is set to the sum of the stages of reusable LFSR from each layer.

B. Graded LFSR Encoding Compression

Encoding the test set based on threshold-based graph-coloring compatible preprocessing can yield an LFSR seed set. During testing, a seed can be decoded via the LFSR to generate multiple deterministic test patterns that are compatible with multiple original test vectors. In the test set obtained after compatible preprocessing, the number of deterministic bits contained in each test vector varies. The more deterministic bits a test vector has, the larger the number of LFSR stages required. Although LFSRs with different stage counts can be obtained in the R3D-LFSR structure, to improve test pattern generation efficiency, LFSRs with fewer stages should be prioritized for test vectors with fewer deterministic bits. Therefore, this paper classifies the test vectors after compatible preprocessing by integrating the test vector generation capabilities of LFSRs available in different test phases.

Assume that the number of LFSR stages on the i -th layer of a stacked n -layer 3D-SIC is denoted as SG_i , where $SG_0 = 0$. The test set is divided into n grades. In the i -th grade, the range of deterministic bits for the test vectors is denoted as $(\sum_{i=1}^{n-1} SG_i, \sum_{i=1}^n SG_i]$. Fig. 8 (d) provides an example of grading. Assume that in the mid-bonding test phase, the test set is for a bonded two-layer die ($n = 2$).

The maximum number of deterministic bits for the test vectors in the first layer is 3, and the first layer has an LFSR stage count of $SG_1 = 3$. The maximum number of deterministic bits for the test vectors in the second layer is 5. Considering that the LFSRs of the two layers can be cascaded, the second layer has an LFSR stage count of $SG_2 = 2$.

The dataset is divided into two grades. The number of test vectors deterministic bits of the first grade ranges from $(0, 3]$ and contains one test vector T_{2+3} . For the first grade, only the first-layer LFSR is used to generate test vectors, and the seed length is SG_1 , so the seed storage size is $1 \times 3 = 3$ bits. The range of test vectors for deterministic bits in the second

grade is $(3, 5]$ and contains three test vectors T_1 , T_{4+6} , and T_5 . The test vectors in the second grade are generated using the longer $LFSR(1+2)$ after cascading of $LFSR1$ and $LFSR2$, with the seed length of $SG_1 + SG_2$, and thus the storage size is $3 \times 5 = 15$ bits. The total storage size required for the test set after the grading strategy is 18 ($3+15$) bits. Without grading, using only the first layer's LFSR, which is too small and has only three stages, does not meet the requirements for generating seeds for T_5 and T_{4+6} . Therefore, it must use the $LFSR(1+2)$ as the test vector generator. At this time, the seed storage is $4 \times 5 = 20$ bits. Hence, the LFSR reseeding scheme using the grading strategy reduces the amount of stored test data.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The experiments are conducted using two groups of benchmark circuits. The first group consists of three-layer 3D-SIC circuits constructed from ISCAS'89 benchmark circuits, where each die integrates two benchmark circuits to emulate a multi-core 3D-SIC architecture. The second group consists of logic-partitioned 3D-IC benchmark circuits constructed from the OpenCores benchmark circuit DES. For each OpenCores circuit, its two-dimensional gate-level netlist is first partitioned into four dies using hMETIS, and the cut nets between different partitions are modeled as inter-layer interconnects such as TSVs. Our first group experiments use circuits from the ISCAS'89 reference circuit and combine them to form a new experimental circuit. The circuit is a three-layer 3D-SIC with Die-1, Die-2, and Die-3, each of which contains two ISCAS'89 benchmark circuits that simulate multi-cores. Table II shows the properties of the experimental circuits.

TABLE II
EXPERIMENTAL CIRCUIT PROPERTIES

Circuit	Reference Circuit	Number of				
		Input	output	flip-flop	Inverter	Gate
Die-1	s1196, s1238	28	28	36	221	816
Die-2	s420, s5378	53	50	195	1853	1144
Die-3	s13207, s15850	62	152	638	5378	2573

To further verify the applicability of the proposed method to logic-partitioned 3D-SICs, this paper adds experiments using a 3D-SIC benchmark circuit constructed from the OpenCores DES circuit. Unlike the 3D-SICs in Table II, which are obtained by combining multiple ISCAS'89 benchmark circuits, the DES benchmark circuit is formed by partitioning a single two-dimensional gate-level circuit and mapping the partitioned logic onto multiple die layers. Therefore, this benchmark circuit can further evaluate the effectiveness and scalability of the proposed test data compression method from the perspective of a logic-partitioned three-dimensional integrated structure.

Table III presents the design details of the logic-partitioned DES-based 3D-SIC. The original two-dimensional gate-level DES circuit is partitioned into four dies, and the cut nets between different dies are modeled as TSVs or vertical inter-layer interconnects. The partitioned DES-based 3D-SIC contains a total of 484 TSVs. The numbers of flip-flops in the four dies are 125, 139, 147, and 134, respectively, and the numbers

of logic gates are 669, 582, 484, and 532, respectively. It can be seen that the partitioning result forms a relatively balanced four-layer 3D-SIC structure, which can be used to evaluate the application effectiveness of the proposed reconfigurable LFSR-based test compression method in multi-layer logic-partitioned circuits.

TABLE III
DESIGN DETAILS OF DES-BASED 3D-SIC BENCHMARK CIRCUIT

Circuit	TSVs	Die1		Die2		Die3		Die4	
		FFs	Gates	FFs	Gates	FFs	Gates	FFs	Gates
DES	484	125	669	139	582	147	484	134	532

For test generation, 10,000 pseudo-random test patterns are first applied to the circuit under test to detect random-pattern-testable faults. Then, a commercial ATPG tool is used to generate deterministic test patterns for the remaining hard-to-test faults. These test patterns are used as the input test set for the proposed compression method. The proposed compression algorithm is implemented in C++, and the proposed method is compared with C-LFSR [17], 3D-LFSR [20], and conventional LFSR reseeding in terms of test data volume, test data storage, area overhead, and test time.

Since this scheme adopts a threshold-based compatibility compression on the original test set before reseeding, Table IV compares the amount of test data for each scheme after the compatibility compression process. The pre-bond test only involves single-layer circuits, the mid-bond test involves the first- and second-layer circuits, and the post-bond test includes first, second, and third-layer circuits. Die-12 refers to the circuit in the mid-bond test, and Die-123 refers to the circuit in the post-bond test. The second column of Table IV gives the Total Number of Test Data (TND) for the original test set. Additionally, the amount of test data (Data After Compression) and compression ratio (CR) for each test phase of the 3D IC are detailed for each scheme. CR is the ratio of the compressed test data volume (or test data storage volume, area overhead, test time) to the original test data volume (or test data storage volume, area overhead, test time). For instance, the CR of the test data volume is

$$CR = \frac{N - N'}{N} \times 100\%,$$

where N is the total number of test data and N' is the data after compression.

TABLE IV
TEST DATA VOLUME AFTER COMPATIBILITY COMPRESSION. THE COMPRESSION RATIO IS IN PERCENTAGE.

	Circuit	Test Data	LFSR	C-LFSR	3D-LFSR	Proposed
Data After Compression	Die-1	768	504	480	480	480
	Die-2	12840	1560	6848	6848	5992
	Die-3	408100	29733	249200	249200	60900
	Die-12	17976	2184	12840	12840	7276
	Die-123	466900	34017	408100	408100	67200
	Total	906584	67998	677468	677468	141848
Compression Ratio	Die-1	768	34.38	37.5	37.5	37.5
	Die-2	12840	87.85	46.67	46.67	53.33
	Die-3	408100	92.71	38.94	38.94	85.08
	Die-12	17976	87.85	28.57	28.57	59.52
	Die-123	466900	92.71	12.59	12.59	85.61
	Total	906584	92.5	25.27	25.27	84.35

The LFSR scheme does not perform compatibility compression. Therefore, the data volume after compatibility compression is the same as the total test data volume. The C-LFSR scheme does not involve compatibility compression. However, it has been processed using the same compatibility compression method as 3D-LFSR to consistently facilitate future performance comparisons of the reseeding method with other schemes. Therefore, the data for both C-LFSR and 3D-LFSR are the same.

Our threshold-based compatibility compression method achieves the highest CR value in all test phases, with a total CR value of 84.35%, significantly better than the other schemes. This is because, in the compatible scheme with no threshold, there may be a few test vectors whose number of deterministic bits increases significantly after compatibility, increasing the maximum number of deterministic bits in the compatible test data set. Although such test vectors are very few, they can cause the S_{max} in the compatible test data set to increase. The number of LFSR stages positively correlates with S_{max} . An increase in the number of LFSR stages increases the seed length, ultimately leading to the inferior compression effect of the compatibility compression method without a threshold compared to our scheme.

Furthermore, the CR value of our scheme on Die-3 and Die-123 both exceeds 85%. This is because the larger the test data size, the more likely it is for the S_{max} in the compatible test set to increase. This indicates that setting a threshold is more meaningful for circuit tests with a larger scale and higher test data volume.

Reseeding the test vectors after compatibility processing to get the seed set. The test data to be stored includes both seeds and control signals. Table V shows the comparison results of the test data storage quantity. In all stages of the test, except

TABLE V
TEST DATA STORAGE CAPACITY. THE COMPRESSION RATIO IS IN PERCENTAGE.

	Circuit	Test Data	LFSR	C-LFSR	3D-LFSR	Proposed
Data After Compression	Die-1	768	504	480	330	330
	Die-2	12840	1560	1830	1344	806
	Die-3	408100	29733	29585	19580	5356
	Die-12	17976	2184	1650	2760	1108
	Die-123	466900	34017	57250	37895	7732
	Total	906584	67998	90795	61909	15332
Compression Ratio	Die-1	768	34.38	37.5	57.03	57.03
	Die-2	12840	87.85	85.75	89.53	93.72
	Die-3	408100	92.71	92.75	95.2	98.69
	Die-12	17976	87.85	90.82	84.65	93.84
	Die-123	466900	92.71	87.74	91.88	98.34
	Total	906584	92.5	89.98	93.17	98.31

for the pre-bond test for Die-1, the compression ratio of this scheme is improved by 3.49%-9.19%. The total compression rate of this scheme is enhanced by 5.14% to 98.31% compared to the 3D-LFSR scheme, which has the second-highest total compression rate, and the total test data storage is reduced by 46,577 bits. The results demonstrate that the threshold-based compatibility compression with hierarchical graded reseeding proposed in this scheme has a strong advantage in improving the test data compression rate. Threshold-compatibility compression reduces the amount of test data needing reseeding, and the patterns are graded according to the number of

deterministic bits contained in mid-bond or post-bond tests. Test vectors with fewer deterministic bits are generated using only the bottom layer *LFSR1*. Test vectors with a medium or large number of deterministic bits are generated, respectively, using the *LFSR(1 + 2)* or *LFSR(1 + 2 + 3)* formed by cascading each layer. This avoids the waste of storage brought about by using LFSR generation of equal length for all test vectors.

Table VI presents the test data volume of the DES-based 3D-SIC after compatibility compression. It can be observed that the proposed method achieves significantly better compression performance than the 3D-LFSR reseeding scheme in all test stages. For the total test data volume of 471,920 bits, the 3D-LFSR method compresses the test data volume to 364,406 bits, with a compression ratio of 22.78 From the

TABLE VI
TEST DATA VOLUME OF DES-BASED 3D-SIC AFTER COMPATIBILITY COMPRESSION. THE COMPRESSION RATIO IS EXPRESSED AS A PERCENTAGE.

	Circuit	TestData	3D-LFSR	Proposed
Data After Compression	Die12	59840	52138	6000
	Die23	66480	48231	7170
	Die34	63120	47131	6540
	Die123	78960	58972	13056
	Die234	95520	73340	14688
	Die1234	108000	84639	19740
	total	471920	364406	67194
Compression Ratio	Die12	59840	12.87	89.97
	Die23	66480	27.45	89.21
	Die34	63120	25.33	89.64
	Die123	78960	25.37	83.46
	Die234	95520	23.22	84.62
	Die1234	108000	21.63	81.97
	Total	471920	22.78	85.76

perspective of different test stages, the proposed method still maintains a high compression ratio in the two-die test stages. Specifically, the test data compression ratios of Die12, Die23, and Die34 are 89.97%, 89.21%, and 89.64%, respectively. In the three-die and four-die test stages, the compression ratio decreases slightly because the circuit scale involved in testing increases, and the number of deterministic test bits and inter-layer interconnect test constraints also increase accordingly. Nevertheless, even in the Die123, Die234, and Die1234 test stages, the proposed method still achieves compression ratios of 83.46%, 84.62%, and 81.97%, respectively. This indicates that the proposed threshold-compatibility compression method can reduce the number of test patterns while suppressing the excessive increase in deterministic bits during compatibility compression, thereby maintaining good test data compression performance.

Table VII further presents the test data storage capacity results of the DES-based 3D-SIC. The test data storage capacity includes the seed data and related control information that need to be stored during LFSR reseeding. For the total test data volume of 471,920 bits, the 3D-LFSR method requires a storage capacity of 162,048 bits, with a compression ratio of 65.66%. In contrast, the proposed method requires only 61,802

bits of storage capacity, achieving a compression ratio of 86.90%. Compared with the 3D-LFSR method, the proposed method further reduces the total test data storage capacity by 61.86%. The test data storage capacity results are basically

TABLE VII
TEST DATA STORAGE CAPACITY OF DES-BASED 3D-SIC. THE COMPRESSION RATIO IS EXPRESSED AS A PERCENTAGE.

	Circuit	TestData	3D-LFSR	Proposed
Data Storage	Die12	59840	18432	5920
	Die23	66480	20736	6816
	Die34	63120	19200	6442
	Die123	78960	28416	11808
	Die234	95520	33792	13536
	Die1234	108000	41472	17280
	Total	471920	162048	61802
Compression Ratio	Die12	59840	69.20	90.11
	Die23	66480	68.81	89.75
	Die34	63120	69.58	89.79
	Die123	78960	64.01	85.05
	Die234	95520	64.62	85.83
	Die1234	108000	61.60	84.00
	Total	471920	65.66	86.90

consistent with the compatibility compression results shown in Table VI. After compatibility compression, the proposed method further classifies the test patterns according to the number of deterministic bits and uses LFSRs of different sizes to perform seed encoding for test patterns in different classes. Therefore, test patterns with fewer deterministic bits do not need to be encoded using an excessively large LFSR. This graded LFSR reseeding strategy avoids the storage waste caused by uniformly encoding all test patterns with a large-scale LFSR. As a result, the proposed method maintains stable storage compression capability across different test stages, with the test data storage compression ratio ranging from 84.00% to 90.11%.

Overall, the experimental results on the logic-partitioned DES-based 3D-SIC show that the proposed method is not limited to the 3D-SIC benchmark circuits constructed by combining ISCAS'89 circuits. Even for a four-layer 3D-SIC constructed by logic partitioning of a single open-source circuit, the proposed threshold-compatibility compression and graded LFSR reseeding methods can still effectively reduce both the test data volume after compatibility compression and the final test data storage capacity. Therefore, the newly added DES-based 3D-SIC experimental results further verify the scalability and generality of the proposed R-BIST architecture and test data compression method in logic-partitioned 3D-SIC testing scenarios.

Table VIII presents the results of the comprehensive analysis performed under the 45nm standard process library for the four reseeding scenarios of the BIST structure. The Area Cost represents the newly added BIST area overhead in this test. In the three pre-bond tests, the area overhead of the other three schemes, including our scheme, sometimes increases and sometimes decreases compared to the Only-LFSR scheme. This is because the scheme reduces the number

TABLE VIII
BIST STRUCTURE AREA OVERHEAD (μm^2)

	Circuit	Only-LFSR	C-LFSR	3D-LFSR	Proposed
Area Cost	Die-1	115.178	122.626	118.636	118.37
	Die-2	151.088	169.442	262.276	147.364
	Die-3	330.638	225.302	359.366	330.638
	Die-12	151.088	169.442	0	0
	Die-123	330.638	234.346	0	0
	Total	1078.63	921.158	740.278	596.372
Compression Ratio	Die-1	115.178	-6.47	-3	-2.77
	Die-2	151.088	-12.15	-73.59	2.46
	Die-3	330.638	31.86	-8.69	0
	Die-12	151.088	-12.15	100	100
	Die-123	330.638	29.12	100	100
	Total	1078.63	14.6	31.37	44.71

of LFSR stages, which reduces the area overhead. On the other hand, the new reseeding scheme adds new coding, decoding, or control structures to the BIST structure, which increases the area overhead. Consequently, both factors work together to show fluctuations in the area overhead. In contrast, the reconfigurable scheme offers a clear advantage in mid-bond and post-bond tests. The non-reconfigurable approach requires adding new LFSRs to the underlying circuits during mid-bond and post-bond tests. For instance, the Only-LFSR scheme adds $151.088\mu m^2$ of hardware area overhead during the Die-12 mid-bond test and $330.638\mu m^2$ of hardware area overhead during the Die-123 post-bond test. The reconfigurable approach adds no new hardware overhead by reusing the pre-bond test's BIST hardware structure in either the mid-bond or post-bond tests. Therefore, during the Die-12 mid-bond test and the Die-123 post-bond test, the newly added BIST area overhead of our scheme is 0. This approach achieves the optimal total compression rate, which is 44.71% compared to the Only-LFSR scheme, and significantly reduces the test area overhead of the 3D-SIC.

TABLE IX
TEST TIME

	Circuit	LFSR	C-LFSR	3D-LFSR	Proposed
Test Time	Die-1	1272	960	810	810
	Die-2	14400	8678	8192	7440
	Die-3	437833	278785	268780	77456
	Die-12	20160	14490	15600	8598
	Die-123	500917	465350	445995	77732
	Total	974582	768263	739377	172036
Compression Ratio	Die-1	1272	24.53	36.32	36.32
	Die-2	14400	39.74	43.11	48.33
	Die-3	437833	36.33	38.61	82.31
	Die-12	20160	28.13	22.62	57.35
	Die-123	500917	7.1	10.96	84.48
	Total	974582	21.17	24.13	82.35

Table IX gives a comparison of the test time. The total test time consists of the test seed and control bit shift-in time, the test vector generation time, and the runtime of the circuit under test. The running time of the circuit under test is not considered here, and the test time is represented by the number of clock cycles consumed in the test. Since this scheme has a favorable compression rate for both test data storage and test data volume, it also excels in test time compression, with an average test time compression rate of 82.35%.

V. CONCLUSION

This paper proposes a reconfigurable R-BIST framework for three-dimensional integrated circuits as a high-performance solution to reduce the BIST test area overhead. The test framework is compatible with the IEEE 1838, 1149.1, 1500, and 1687 standards and can perform serial or parallel tests, which is conducive to reducing the test time. The test data compression method combines threshold-compatibility compression with graded LFSR encoding compression, and reduces the amount of test data and data storage. This scheme takes into account both the characteristics of the hardware structure and the test data.

The effectiveness of the proposed method is demonstrated with the results tested using simulated 3D-SIC from the IS-CAS'89 circuits. The area overhead of the BIST test structure is reduced by 44.71%, the amount of test data volume is reduced by 84.35%, the amount of test data storage is reduced by 98.31%, and the test time is reduced by 82.35%. This leads to a significant reduction in the test cost. Moreover, this scheme is more effective in compressing large-scale circuits.

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